

Printer and Scanner

Daisy-Wheel Printers

- works on the same principle as ball-head typewriter
- consists of a disk made of plastic or metal on which characters stand out along the outer edge.
- The printer rotates the disk to print a character until the desired letter is facing the paper, after which a hammer called solenoid strikes forcing the character to hit an ink ribbon making a mark of the character on the paper.
- speed is rated by cps (number of characters per second).
- main disadvantage is that they make noise when printing and these kind of printers cannot print graphics.
- advantage is that they are not expensive and can produce letter-quality text.



Dot-Matrix Printers

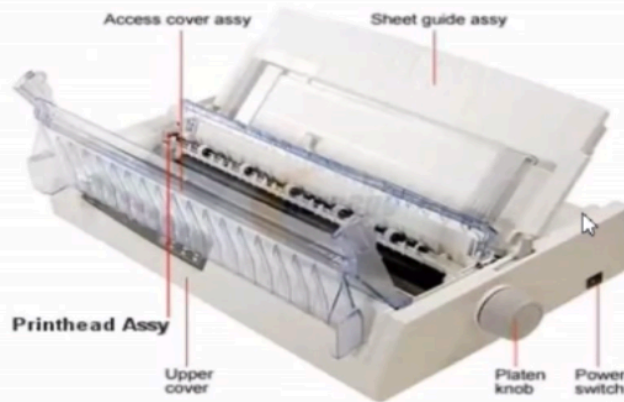


Fig.:
Continuous
feed paper

Fig.: Dot-matrix printer

Figure 1: Dot Matrix

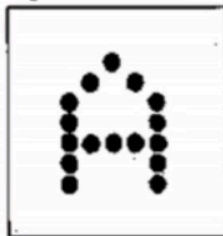


Figure 2: Print Head

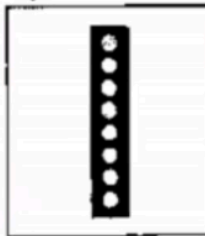


Figure 3: Descenders

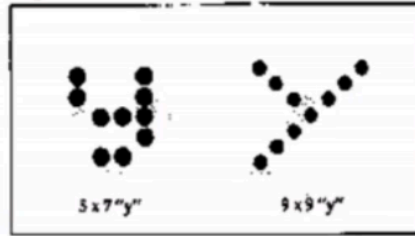


Figure: 8-pin dot-matrix print head with examples of printed letters.

Dot-Matrix Printers

- **Advantages of Dot-matrix printers include the following:**
 - They can print on multi-part stationary or carbon copies
 - Lower printing costs compared with Inkjet or Laser, the reason being that dot-matrix printers use a ribbon and rugged printing process.
 - These can withstand unclean or dusty environment whereas Inkjet or laser jet printers require clean environment
 - Suitable for traction fed paper or continuous paper feed. Sometimes you may need to print an activity using continuous paper feed. Inkjet and Laser printers use discrete sheets of paper, and normally do not use continuous paper feed.
 - Using these printers require negligible operator training.
 - These printers usually cost less (initial purchase cost may be more, but running cost is negligible)
- **The main disadvantages of Dot-matrix printers are:**
 - Dot-matrix printers are noisy but not as much as Daisy-wheel printers
 - The quality of print is not as good as laser and inkjet printers.
 - cannot be used for producing good quality images or for photo printing.
 - These printers are slow compared with laser printers.

Solid-Ink Printers

- Solid-Ink printers use ink in a waxy solid form than liquid form which avoids problems like spillage.
- And these kinds of printers print one line at a time and these printers are best suitable for graphic companies that need true color at a price lower than a color laser printer.
- **Advantages and Disadvantages**
 - Advantages are good print quality, ease of use and generate less waste
 - Solid ink printers are less sensitive to paper thickness and paper fibers. This allows printing on a number of different surfaces including recycled and handmade paper.
 - disadvantages include more power consumption and odour of wax

Thermal Printers

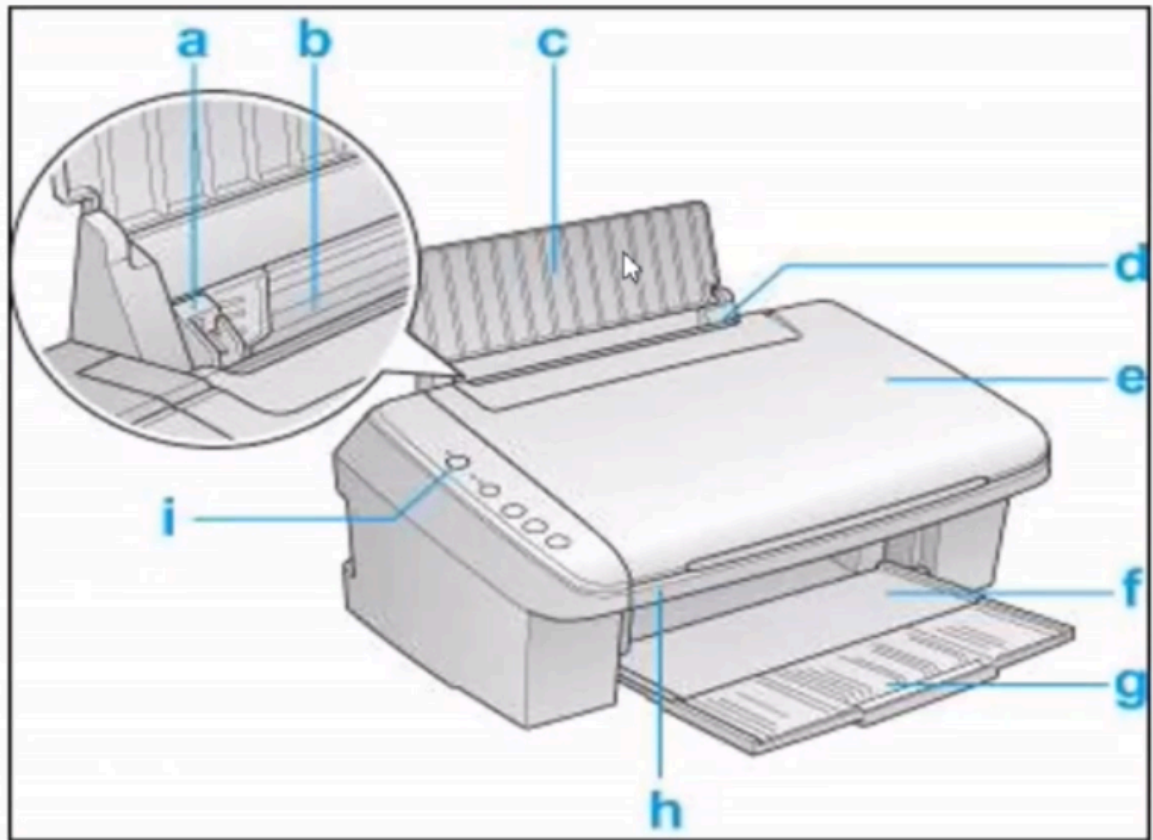
- **Direct Thermal printer**

- Direct Thermal printer use a heated printhead to burn dots into the surface of special heat-sensitive paper.
- It is similar to older fax machines. The disadvantage in this is that the paper that is used gets darken early, thus making difficult to read.

- **Thermal wax-transfer printer**

- Thermal wax-transfer printers use a heat-sensitive-ribbon instead of heat-sensitive paper.
- Thermal printhead melts wax-based ink from the ribbon on to the paper.
- The disadvantages with thermal transfer printers are that the heating and cooling of the print head determines their speed which means that since the printhead is extensively used, it has to be replaced often.

Inkjet Printers



Inkjet Printers

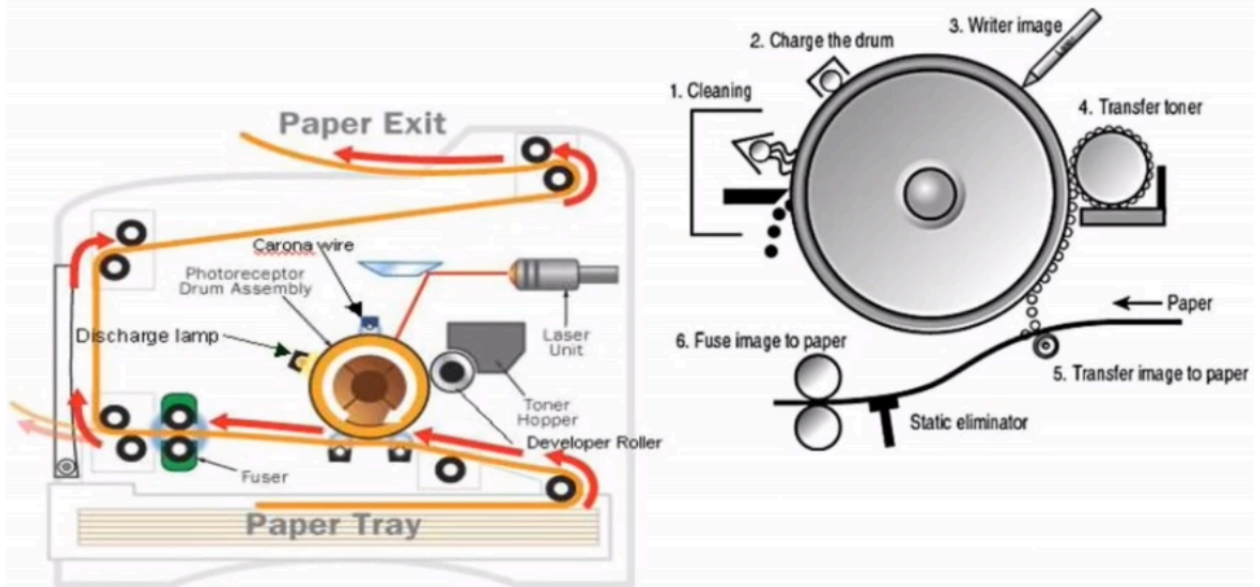
- a) Edge guide - Helps load the paper straight. Adjust the left edge guide so that it fits snugly to the width of your paper.
- b) Sheet feeder - Feeds a stack of paper automatically.
- c) Paper support - Supports the paper loaded in the sheet feeder.
- d) Feeder guard - Prevents objects placed on the document cover from falling inside the printer when opening the document cover.
- e) Document cover - Open and close when you place a photo or document.
- f) Output tray - Receives ejected paper.
- g) Output tray extension - Supports the ejected paper.
- h) Scanner unit - Open and close when you replace an ink cartridge.
- i) On button - Turns the printer on and off.

Inkjet Printers

- **Advantages and Disadvantages**

- The low cost and relatively high quality of prints that are offered by the inkjet printer is suitable for most day-to-day tasks and thus used in home and offices.
- Inkjets have the advantage of practically no warm up time and lower cost per page, no noise
- Inkjet printers can print on a variety of papers as they essentially work by shooting printer ink on to the paper.
- the disadvantages are ink cartridges are expensive.

Laser Printer



Laser printer components and their functionality

Laser Printer

1. Paper feeding

- The printer moves a sheet of paper from the proper tray onto a series of rollers, through the imaging and fixing areas, and to the output hopper.

2. Drum Cleaning and Charging

- Any residual toner from past jobs is scraped from the printer's photosensitive drum. A fine wire (the primary corona) produces a negative electrical charge across the entire face of the drum. The image is set in raster lines as a series of fine dots on the drum.

3. Imaging the Drum

- The information from the raster-image processor is read from memory and sent to the print engine, one line at a time. The laser sets a positive charge in the areas of the image to be filled with toner.

4. Transferring Toner to the Drum

- A film of fine plastic power is placed on the toner transfer roller, which is turning close to the photosensitive drum. This toner is then attracted to the positively charged areas of the drum.

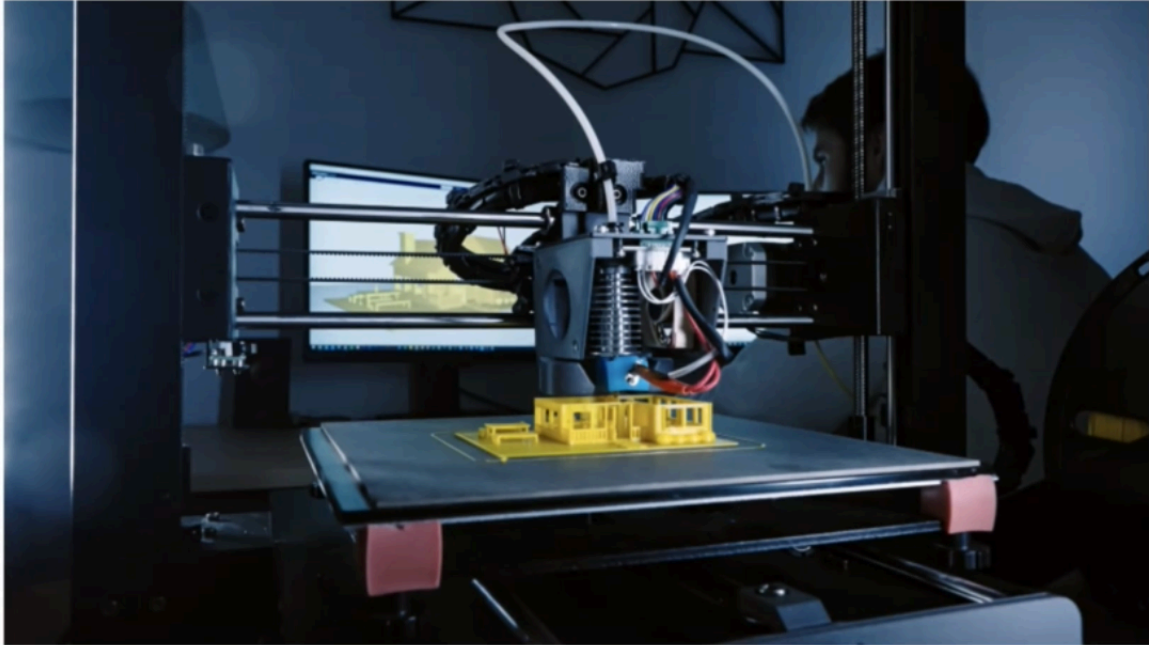
5. Transferring Toner to the Paper

- The corona wire places a positive electrical charge on the paper as it moves close to the drum. The toner is attracted to the page, forming an image.

6. Fusing the Toner

- The page passes through a pair of rollers. The roller on the side toward the toner that has been placed on the page is heated just enough to melt the plastic toner particles onto the page without smearing. The roller on the other side supplies the needed pressure.

3D Printer



3D PRINTING USES

- Construction
- Automotive
- PPE
- Medical Equipment
- Food
- Aerospace
- Education

The 3D Printing Process

- 3D printers are related to additive manufacturing. 3D printers use computer-aided design to understand a design. When a design is ready, a material that can be dispensed through a hot nozzle or precision tool is printed layer by layer to create a three-dimensional object from scratch.
- When the modeling and slicing of a 3D object is completed, it's time for the 3D printer to finally take over. The printer acts generally the same as a traditional inkjet printer in the direct 3D printing process, where a nozzle moves back and forth while dispensing a wax or plastic-like polymer layer-by-layer, waiting for that layer to dry, then adding the next level. It essentially adds hundreds or thousands of 2D prints on top of one another to make a three-dimensional object.

3D Printer

- **HOW MUCH DO 3D PRINTERS COST?**
 - The cost of 3D printers vary based on the size, specialty and use. The cheapest 3D printers, for entry level hobbyists, typically range from \$100 to \$500. More advanced models can range between \$300 and \$5,000. Industrial 3D printers can cost up to \$100,000.

3D Printing

- **Advantages**

- 3D Printers are Affordable
- 3D Printers are Fast
- 3D Printers Can Work with Speciality Materials

- **Disadvantages**

- 3D Printers May Not Provide Enough Strength
- 3D Printers May Have Accuracy Issues
- 3D Printers May Require Post-Processing



Introduction to Scanners

- Scanner is a device that scans images, printed text, and handwriting etc and converts it to digital form or image.
- It is so named because the data is converted one line at a time or scanned down the page as the scanning head moves down the page.

Components inside a scanner

- Glass Plate and Cover

- The glass plate is the transparent plate wherein the original is placed so that the scanner can scan it and the cover keeps out stray light that can affect the accuracy of the scan

- Stepper Motor

- The stepper motor in a scanner moves the scan head down the page during scan cycle and this is often located either on the scan head itself or attached to a belt to drive the scanner head.

- Scanning head

- Scanning head is the most important component because it is the one which does actual scanning. It contains components like
- **Light source and mirror** : It is the bright white light that is used to illuminate the original as it is being scanned and which bounces off the original and reflected off several mirrors

Components inside a scanner

- **Stabilizer bar:** It is a long stainless steel rod that is securely fastened to the case of the scanner and it provides a smooth ride as the scanner scans down the page
- **CCD (Charge Coupled Device) or CIS (Contact Image Sensor):** A CCD array is a device that converts photons into electricity. Any scanner that uses CCD use lens to focus the light coming from the mirrors within the scanning head.
- Another technology used in some cheaper scanners is CIS wherein the light source is a set of LEDs that runs the length of the glass plate.

Flatbed Scanners



- The most commonly used scanner is a flatbed scanner also known as desktop scanner.
- It has a glass plate on which the picture or the document is placed.
- The scanner head placed beneath the glass plate moves across the picture and the result is a good quality scanned image.
- For scanning large maps or toposheets wide format flatbed scanners can be used.

Sheetfed Scanners



- Sheet fed scanners work on a principle similar to that of a fax machine.
- In this, the document to be scanned is moved past the scanning head and the digital form of the image is obtained.
- The disadvantage of this type of scanner is that it can only scan loose sheets and the scanned image can easily become distorted if the document is not handled properly while scanning.

Handheld Scanners



- Hand-held scanners although portable, can only scan images up to about four inches wide.
- They require a very steady hand for moving the scan head over the document.
- They are useful for scanning small logos or signatures and are virtually of no use for scanning maps and photographs.